

Course Map

Combustion-driven, PVC, adult-supervised

Four lessons and a build, on open ground

A combustion potato launcher is a heat engine that runs once. Fuel and air burn in a closed chamber, the gas gets hot, hot gas pushes hard, and the only thing free to move is the potato. Every lesson here is one term in that sentence, and each one is measured rather than asserted — because the difference between a launcher that works and one that bursts is entirely a matter of numbers you can check beforehand.

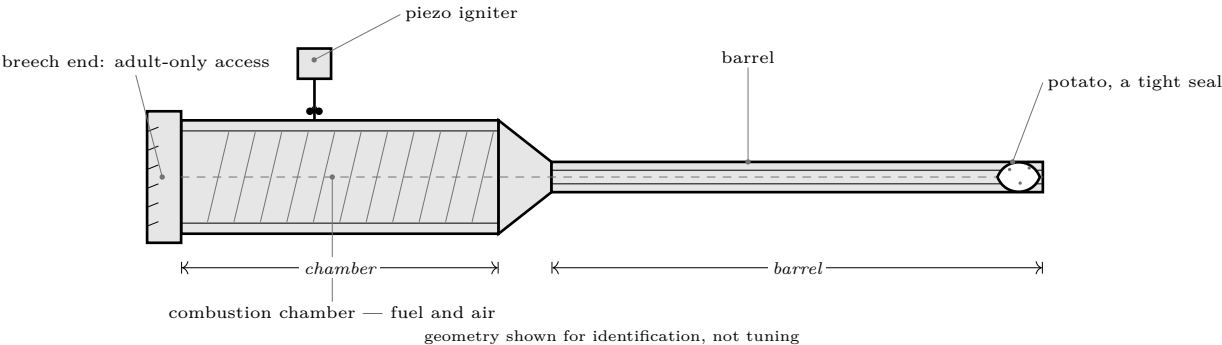
THIS IS A PROJECTILE WEAPON

A combustion launcher throws a half-pound potato several hundred feet at speeds that will kill a person. It is not a toy, it is not a firework, and the fact that it is made of plumbing parts changes nothing about what it does downrange. Every rule on the **Safety** sheet applies every time: adult present and in charge, muzzle discipline as strict as a rifle's, a confirmed clear range, eyes and ears on, and never a person, animal, building, road or anything you cannot afford to destroy in the direction it is pointing.

The through-line

#	Lesson	The idea	Why the launcher needs it
1	Air is stuff	Air has mass, exerts pressure, and pushes in every direction at once. Crush a can with it.	The chamber is full of it, and it is half the fuel.
2	Combustion	Flame needs fuel, oxygen, and ignition; mixtures can be lean or rich.	Explains why adding fuel after a misfire is forbidden, not how to choose a dose.
3	Gas laws	Heat can change gas pressure, but an ideal model cannot certify a chamber.	Separates prediction from approval.
4	Projectile motion	Range, drag, and uncertainty.	Shows why an equation never defines a safe boundary.
5	Verify, then decide	Inspect, document, apply every stop gate.	Missing evidence means no shot.

The machine



The craft track — run this alongside the lessons

The lessons explain why the launcher works. The craft sheets teach how to build it *well*, and they are the part that transfers to everything else either of them ever makes.

#	Craft sheet	The standard it sets
1	Measure & Mark	One thin line, drawn square all the way round, from a measurement taken twice.
2	Cut Square	Perpendicular within a degree. No light under a try square, checked in four positions.
3	Deburr & Chamfer	No burr anywhere; an even 1–2 mm chamfer on every pipe end.
4	Solvent Weld	A continuous squeeze-out bead all the way round, pipe bottomed, witness mark aligned, date written on the part.
5	Drill & Seal	Round holes, no crazing, epoxy fillet continuous — and the spark tested before anything is sealed.
6	Inspect Your Own Work	Ten named checks, run before the step that hides the evidence, fixed rather than accepted.
—	Build Log	The written record that turns “I think it cured” into a date and a signature.

Why the craft track exists

A launcher is never certified safe by these pages. Conservative materials, sound work, and disciplined operation reduce risk; none removes it. Care varies with mood, tiredness and how many people are waiting. A method with written checkpoints does not. That is the whole reason trades, laboratories and hangars work this way, and it is worth more than the launcher.

How each lesson runs

Predict first	Every lesson opens with a question and a box to commit an answer in, before anything is lit. A wrong prediction written down teaches far more than a demonstration nobody doubted.
Close each phase	Stop the apparatus, isolate every energy source, let hot parts cool, account for materials, and have the adult say the bench is reset before discussion or the next activity begins.
They build it	The boys cut, deburr, dry-fit, glue and assemble. The adult owns fuel, ignition and the decision to fire.
Measure it	Every lesson produces a number: a pressure, a ratio, a distance, a velocity. Numbers are what separate this from a stunt.
Challenge the model	Change one variable only in a bench model or calculation. Never vary fuel, projectile, or launcher geometry to find a limit.
Then the maths	The equation arrives as the compact way to state what they already measured.

Who does what

Lessons 1–4	Bench work: a can, a candle, a balloon, a tape measure. Nothing here is dangerous beyond ordinary kitchen care.
The build	The boys do essentially all of it. Cutting, deburring, solvent welding and the 48-hour cure are the bulk of the work and none of it involves fuel.
Firing	The adult is in charge of the fuel can, the igniter and the command to fire, permanently. Everyone else is behind the firing line.
Where	Only a walked, controlled site with a positive earthen backstop and no road, building, trail, animal, boundary, or unseen ground in the line of fire. A predicted distance is never a boundary.
Before anything	Read the Safety sheet start to finish, and check local ordinances — these are legal in most places and restricted in some, and that is on you to confirm before you build.

Pressure-rated pipe only. Schedule 40 *pressure* PVC, printed with an NSF-PW rating and a pressure figure. Never DWV, never foam-core, never cellular-core, never thin-wall drain pipe — those are not rated for internal pressure at all and they fail by shattering into fragments rather than splitting. This is the one specification in the whole project that is

not negotiable.

The one sentence version

A permitted, documented fuel system may burn rapidly; the resulting pressure acts on the base of the potato all the way down the barrel; and the potato leaves the muzzle carrying the energy the fuel released. Lessons 1 to 4 are the four clauses of that sentence.

The evidence contract

Claim	Evidence required
Site usable	Permission and law checked; range walked; earthen backstop and boundaries positively identified.
Assembly eligible	Markings recorded; cure derived from the actual cement instructions and conditions; no reject finding.
Session controlled	Named adult lead and watcher; marked line; PPE; spoken states and cease-fire understood.
Result understood	Conditions and anomalies logged. An unexplained result closes the session; it does not invite another shot.

ACCEPTANCE GATE

Missing evidence is a failed gate. No estimate, classroom equation, prior success, or quick test substitutes for it.

Decision practice

Observation	Disposition and evidence
The site was clear last week	Walk it again. Record today's boundaries, backstop, weather, fire status, and permission.
A joint looks sound but its assembly time is missing	Reject it. Appearance cannot reconstruct cement, temperature, humidity, or elapsed cure time.
The first result is weak or sounds different	Call cease-fire, unload and vent when the state permits, log the anomaly, and end the session. Do not tune.
The calculation predicts a short landing	Ignore it as a boundary. Confirm the whole walked range and positive earthen backstop.
Someone is unsure which state was called	Stop and return to cold, open, and vented before discussion. Uncertainty is itself a stop condition.

Review aloud

Who may call cease-fire? What evidence makes a cure record complete? What is the difference between a pressure marking and combustion approval? Which finding retires a part rather than inviting repair? Why is a successful prior shot not evidence for the next one?

Course acceptance. Each learner can identify the cold state, name the adult-only actions, trace the misfire hold without prompting, distinguish a model from a boundary, and point to the written record behind every proceed decision.

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